

A Design Proposal for Mach-Zehnder Interferometer

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Index Terms—Silicon photonics, Mach-Zehnder interferometer.

I. INTRODUCTION

The objective of this project is to design, simulate, fabricate, and experimentally characterize silicon photonic interferometers on a 220-nm silicon-on-insulator (SOI) platform.

The primary device investigated in this proposal is an unbalanced Mach-Zehnder interferometer (MZI). The interferometer will be used to study the spectral response of silicon waveguides and to experimentally extract the waveguide group index from the measured free spectral range (FSR).

The design study considers variations in parameters including the interferometer path-length difference, $\Delta L = L_2 - L_1$. The simulated and measured results will be compared to evaluate the accuracy of the waveguide and circuit models as well as fabrication-induced variations.

II. WAVEGUIDE DESIGN

A. Waveguide Geometry

The devices are implemented using strip waveguides on a silicon-on-insulator platform. The nominal waveguide parameters are summarized in Table I.

TABLE I
NOMINAL WAVEGUIDE DESIGN PARAMETERS.

Parameter	Value
Silicon thickness	220 nm
Waveguide width	500 nm
Polarization	quasi-TE
Nominal wavelength	1550 nm
Waveguide type	Strip waveguide
Cladding	SiO ₂

B. Simulated Optical Mode

The optical mode of the waveguide is simulated using Lumerical MODE. The fundamental quasi-TE mode is used for the device design.

Figure 1 shows the simulated modal field distribution.

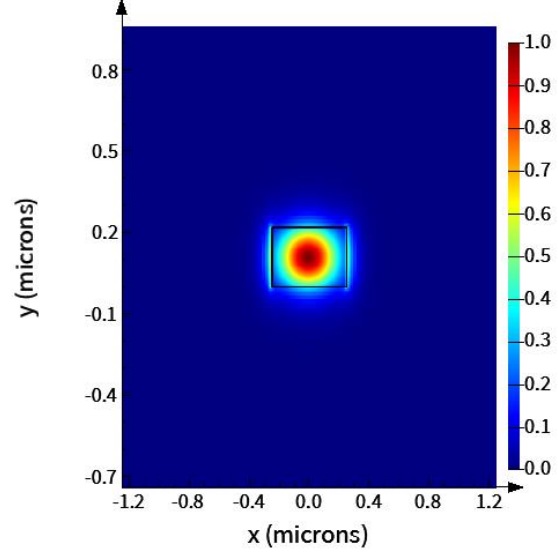


Fig. 1. Simulated fundamental quasi-TE mode of the 500-nm-wide, 220-nm-thick silicon strip waveguide at $\lambda = 1550$ nm.

At the nominal wavelength, the simulated effective index is

$$n_{\text{eff}}(\lambda_0) = 2.5006. \quad (1)$$

III. EFFECTIVE INDEX AND GROUP INDEX

The wavelength dependence of the effective index is obtained from Lumerical MODE over the wavelength range 1500 - 1600 nm. The group index is defined as

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (2)$$

Figure 2 and figure 3 show the simulated effective and group indices.

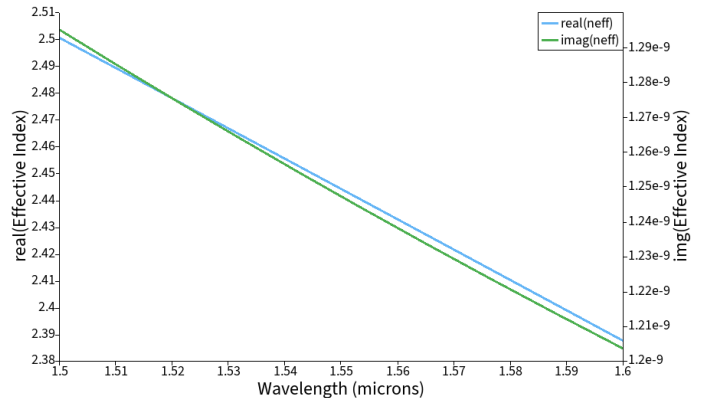


Fig. 2. Simulated effective index n_{eff} versus wavelength.

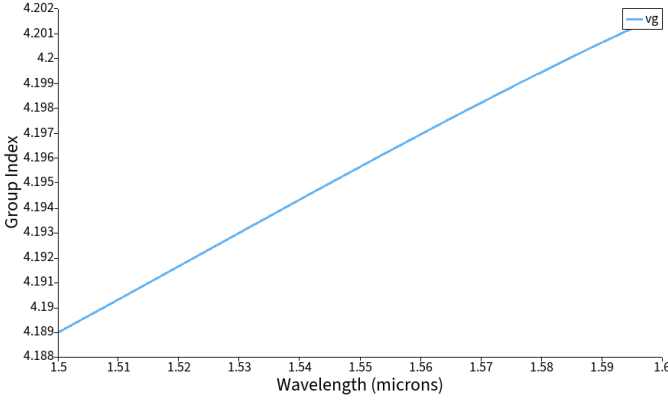


Fig. 3. Simulated group index n_g versus wavelength.

At $\lambda_0 = \text{nm}$, the calculated group index is

$$n_g(\lambda_0) = 4.189. \quad (3)$$

IV. WAVEGUIDE COMPACT MODEL

The wavelength-dependent effective index is represented by a polynomial compact model.

Defining

$$\Delta\lambda = \lambda - \lambda_0, \quad (4)$$

the effective index is approximated as

$$n_{\text{eff}}(\lambda) = a_0 + a_1\Delta\lambda + a_2(\Delta\lambda)^2. \quad (5)$$

The fitted coefficients are listed in Table II.

TABLE II
POLYNOMIAL COMPACT-MODEL COEFFICIENTS FOR THE WAVEGUIDE.

Coefficient	Value
a_0	2.44682
a_1	-1.133
a_2	-0.0439

V. MACH-ZEHNDER INTERFEROMETER DESIGN

A. Device Structure

The primary circuit is an unbalanced Mach-Zehnder interferometer. The input optical field is divided into two arms by a splitter. The arms have lengths L_1 and L_2 , with path-length difference

$$\Delta L = L_2 - L_1. \quad (6)$$

The optical fields are subsequently recombined at the output splitter. The nominal splitter type used in this design is the Y-branch.

B. MZI Transfer Function

For an ideal lossless 50:50 interferometer, the fields in the two arms acquire phases

$$\phi_1 = \beta L_1, \quad \phi_2 = \beta L_2, \quad (7)$$

where

$$\beta(\lambda) = \frac{2\pi n_{\text{eff}}(\lambda)}{\lambda}. \quad (8)$$

The relative phase difference is therefore

$$\Delta\phi = \phi_2 - \phi_1 = \beta(\lambda)\Delta L = \frac{2\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda}. \quad (9)$$

For ideal 50:50 splitting and recombination, one output has normalized power transmission

$$T(\lambda) = \frac{1}{2} [1 + \cos(\Delta\phi)]. \quad (10)$$

Equivalently,

$$T(\lambda) = \cos^2 \left[\frac{\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda} \right]. \quad (11)$$

Depending on the selected output port and splitter phase convention, the complementary output may instead follow

$$T_{\text{comp}}(\lambda) = \sin^2 \left[\frac{\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda} \right]. \quad (12)$$

VI. FREE SPECTRAL RANGE AND GROUP INDEX

Constructive interference occurs when

$$\Delta\phi = 2\pi m, \quad (13)$$

where m is an integer.

Using Eq. (9),

$$\frac{2\pi n_{\text{eff}}(\lambda)\Delta L}{\lambda} = 2\pi m. \quad (14)$$

Thus,

$$m = \frac{n_{\text{eff}}(\lambda)\Delta L}{\lambda}. \quad (15)$$

For adjacent interference orders, the change in m is one. Differentiating with respect to wavelength gives

$$\frac{dm}{d\lambda} = \Delta L \frac{d}{d\lambda} \left(\frac{n_{\text{eff}}}{\lambda} \right). \quad (16)$$

Expanding the derivative,

$$\frac{dm}{d\lambda} = \Delta L \left[\frac{1}{\lambda} \frac{dn_{\text{eff}}}{d\lambda} - \frac{n_{\text{eff}}}{\lambda^2} \right]. \quad (17)$$

Using the definition

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad (18)$$

we obtain

$$\frac{dm}{d\lambda} = -\frac{n_g \Delta L}{\lambda^2}. \quad (19)$$

Therefore,

$$\Delta\lambda_{\text{FSR}} \approx \frac{\lambda^2}{n_g \Delta L}. \quad (20)$$

Rearranging gives the group index that can be extracted from the measured spectrum:

$$n_g \approx \frac{\lambda^2}{\Delta L \Delta\lambda_{\text{FSR}}}. \quad (21)$$

Equation (21) will be used to extract the experimental group index from the fabricated MZI.

VII. DESIGN VARIATIONS

Multiple devices will be fabricated in order to investigate the dependence of the spectral response on waveguide geometry and interferometer path-length difference.

Table III summarizes the proposed devices.

TABLE III
PROPOSED DEVICE VARIATIONS.

Device	w (nm)	Pol.	ΔL (μm)	n_g	FSR (nm)
MZI-1	500	TE	100	4.189	5.735
MZI-2	500	TE	50	4.189	1.147
MZI-3	500	TE	800	4.189	0.717

The ΔL variations are designed to test the predicted inverse relationship

$$\text{FSR} \propto \frac{1}{\Delta L}. \quad (22)$$

VIII. CIRCUIT SIMULATION

The complete photonic circuit is simulated using Lumerical INTERCONNECT (Figure 4) using the compact models for waveguides, grating couplers, and splitters.

Figure 5 shows the predicted transmission spectrum of one representative interferometer with a path length difference of $100 \mu\text{m}$.

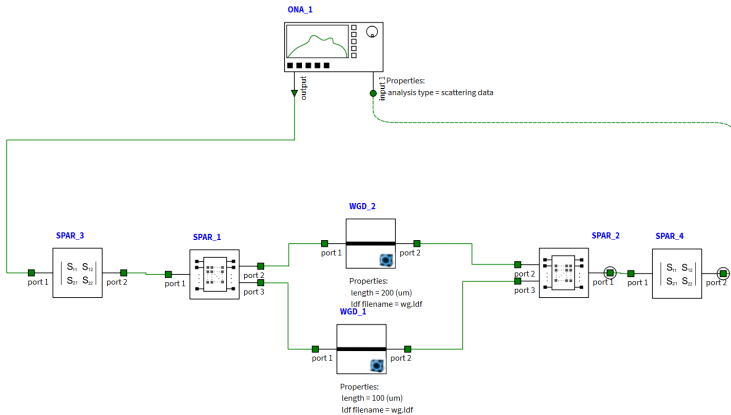


Fig. 4. The MZI circuit implemented in INTERCONNECT.

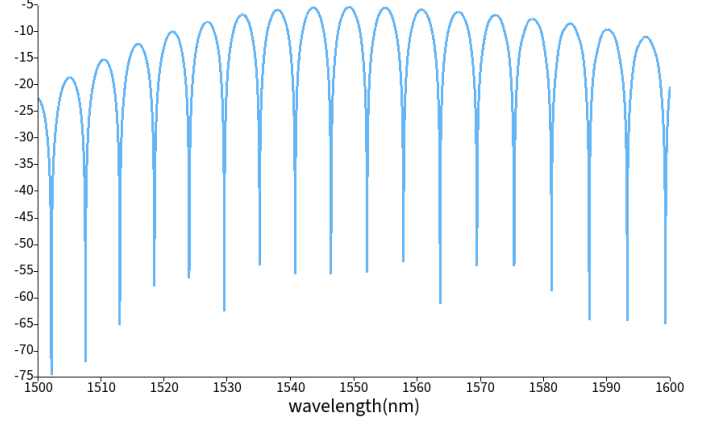


Fig. 5. Simulated transmission spectrum of the MZI circuit with a path difference of $100 \mu\text{m}$.

From the simulated spectrum,

$$\text{FSR}_{\text{sim}} = 5.68 \text{ nm}. \quad (23)$$

Using Eq. (21), the group index extracted from the simulated MZI spectrum is

$$n_{g,\text{MZI}} = 4.23. \quad (24)$$

This result agrees with the group index calculated directly from the MODE waveguide simulation.

IX. CONCLUSION

This project investigates silicon photonic interferometers fabricated on a 220-nm SOI platform. The design uses unbalanced interferometers with controlled path-length differences to generate measurable spectral fringes.

The free spectral range provides a direct experimental method for extracting the waveguide group index. By comparing MODE simulations, compact-model circuit simulations, and experimental measurements, the project will evaluate both the optical design and the accuracy of the waveguide model.